From a Cold Plate to a Melt Pool

Everything built and verified so far, the one measurement that fails, and what happens next

10 September 2026 | supersedes roadmap rev. 4 (1 Sep) | Al–5%Mg, solidus 815 K, liquidus 906 K | OpenFOAM ESI v2506

THE STATE IN FIVE LINES

A weld pool now exists. Continuous above the solidus from the second droplet, continuously liquid from the sixth, with sustained convection. Until 9 September this project had never produced a single cell of liquid metal.

Mass, energy, momentum and the domain all check out. Bead cross-section lands 0.38% from a target fixed before the run; energy balances; the far field never moves off 300 K; solid metal holds velocity at exactly zero.

One number fails. The bead is **2.60 mm** tall where Zhao measures 1.55 mm. The replication gate is 15%; this is +68%.

The cause is diagnosed, not mysterious. Two forces present in Zhao's model are absent from ours — arc pressure (228 Pa) and the welding current's self-field (435 Pa) — and they are precisely the forces that flatten and spread a bead.

Nothing on the critical path needs the shared server. The next two solver edits and the run that tests them fit on the desktop.

This document does three things. It explains the background behind every mechanism in the model, so that no equation or code fragment appears without the reasoning that put it there. It then walks the work in the order it happened, including the things that were wrong and how they were found. It closes with the plan. Where a claim rests on a measurement, the measurement is given; where it rests on an argument, the argument is written out.

1. The thesis, and why it is difficult

1.1 What Wire Arc Additive Manufacturing is

Wire Arc Additive Manufacturing (WAAM) builds a metal part by welding it into existence. A welding torch traverses a substrate; an electric arc struck between a continuously fed wire electrode and the workpiece melts both. The wire melts into droplets that detach, fall through the arc, and land in the pool of molten metal the arc has created. The torch moves on, the pool solidifies behind it, and what is left is a bead of deposited metal. Repeat the pass on top of itself and a wall grows; plan the paths and a part grows. The specific process here is **GMAW** — Gas Metal Arc Welding — in which the consumable wire is itself the electrode and a shroud of argon keeps the atmosphere off the molten metal.

The attraction is deposition rate and cost: kilograms per hour from a welding power supply, against grams per hour from a powder-bed laser machine costing twenty times more. The problem is the microstructure. Metal that solidifies from a large, slowly cooling weld pool grows **coarse columnar grains** aligned with the heat-flow direction. Coarse columnar grains make the part anisotropic — strong along the build direction,

weaker across it — and they are where cracks start. Fine equiaxed grains would be better in every respect. Getting them is the central materials problem of WAAM.

1.2 Why a magnetic field

Molten aluminium is an electrical conductor, and during welding it carries the welding current — 135 A in this study. A current-carrying conductor in a magnetic field experiences the **Lorentz force**, **J** x **B**, where **J** is current density and **B** is magnetic flux density. Impose an external field across the pool and you add a body force to the liquid metal that was not there before. That force stirs the pool.

Stirring matters because grain structure is set by what happens at the solidification front: the local thermal gradient G, the growth rate R, and whether fragments of already-formed dendrite break off and float away to seed new grains. Forced convection changes all three. The experimental literature consistently reports grain refinement under an applied field. The mechanism is argued about — thermal-gradient modification versus dendrite fragmentation — and that argument is what a simulation can address, because a simulation can separate effects that an experiment can only observe together.

THE DISTINCTION THAT DEFINES THIS PROJECT

Two entirely different couplings both get called "the Lorentz force", and the difference decides whether the thesis has a subject at all.

Braking arises when a conductor moves through a field: the motion induces a current, **J** = sigma(**u** x **B**), and the resulting force opposes the motion. It scales as B^2 and it always damps. It requires no current source — motion is enough.

Stirring arises when an externally supplied current crosses a field. It scales linearly in B, and it *drives* flow rather than damping it. It requires a current source: here, the 135 A welding arc.

At the melt pool the arc's current density is about **1.07e7 A/m^2**. Comparing the two as stresses over the pool scale, stirring beats braking by **491x at 20 mT** and **196x at 50 mT**. They cross only at 9.8 T, a field no welding experiment applies.

The solver as shipped has only the braking term, because it was written for lasers, where there is no welding current. Injecting the current is therefore the core technical contribution of this thesis and not an optional refinement.

1.3 The replication target

Before adding a magnetic field to a model, the model has to be right without one. The reference is **Zhao, Wei, Liu, Long, Chen and Ou (2021), *Welding in the World* 65:1571–1590**: a three-dimensional simulation of GMAW-based WAAM in Al–5%Mg that couples droplet formation, detachment and impingement to the molten pool, validated against their own experiments. Deposit height and length agree with measurement to within 15%, and for their four-layer case to 4.3% in height. Their process parameters — 135 A, 16 V, 120 cm/min travel, 900 cm/min wire feed, 1.2 mm wire — are the ones used here. Their Table 4 supplies the material properties, Table 5 the dimensions we must reproduce.

There is no magnetic field anywhere in that paper. That is the point: it is a clean, validated, field-free baseline. Reproduce it, then switch the field on, and any difference is attributable to the field rather than to the dozen other things a new model gets wrong.

1.4 Why this is a fluid-dynamics problem, not a heat-conduction one

It would be convenient if a weld pool could be predicted by solving the heat equation with a moving source. It cannot, for three reasons that all involve fluid motion.

First, the pool convects. Liquid metal in the pool moves at up to a metre per second, and moving metal carries heat with it far faster than conduction does. Second, the free surface deforms. Droplets arrive at 0.9 m/s and dent it; arc pressure pushes it down; surface tension pulls it back. The shape of that surface *is* the bead geometry, which is the thing being predicted. Third, the pool has a free boundary between metal and argon that moves, merges and reconnects as each droplet lands. All three demand a two-phase flow solver with a moving interface, phase change, and surface forces.

2. The toolchain, and the reasoning behind each choice

2.1 OpenFOAM, and the fork question

OpenFOAM is an open-source C++ library for computational continuum mechanics, distributed as a set of solvers built on shared discretisation machinery. It matters here that it is open: this project modifies the solver, and that is only possible with source access.

OpenFOAM exists as **two incompatible forks** with a shared ancestor and diverged APIs: the **Foundation** line (openfoam.org, versioned 9, 10, 11) and the **ESI** line (openfoam.com, versioned by date, v2506 = June 2025). Code written against one does not compile against the other. Phase 1 settled on **ESI v2506**, and the reason is in the next section.

2.2 laserbeamFoam and the magnetic module

`laserbeamFoam` is a research solver for laser melting that already couples the four things this problem needs: volume-of-fluid interface tracking, enthalpy-porosity melting with latent heat, Continuum Surface Force surface tension with a Marangoni term, and Boussinesq buoyancy. Building that from scratch would be most of a thesis on its own.

Its ESI mainline branch also ships `src/MTHD`, a magnetothermohydrodynamics module: an induction equation for the magnetic field **H** with divergence cleaning, **J** = curl(**H**), a **J** x **B** term already present in every solver's momentum equation, Joule heating, and electrical conductivity blended across the phase interface. It comes with a Hartmann-flow validation tutorial. That module was the highest-risk item in the original plan and it turned out to already exist — which is why the ESI fork was chosen over the Foundation one, which has no MHD at all.

THE PHYSICS LEDGER, SETTLED IN PHASE 1

Ten items ship free and need no work: VOF, enthalpy-porosity, latent heat, CSF surface tension, Marangoni, Boussinesq buoyancy, Lorentz force, Joule heating, conductivity and viscosity blending across the interface, and an applied external field through boundary conditions on **H**.

Six are self-contained additions that do not require restructuring the solver: the Goldak arc heat source, arc pressure, plasma drag, surface heat loss, the moving wire inlet, and the annular shielding-gas inlet.

One is genuinely new code: injecting the 135 A welding current, so that **J** is the arc's current and not merely the motional sigma(**u** x **B**) that a laser code provides.

2.3 The Edit numbering

Changes to the solver and the case are numbered as **Edits** so that any result can be traced to the exact state of the code that produced it. Edits 1 to 8 covered the arc heat source, the temperature-dependent property model, and boundary-condition defects; Edit 4 is the droplet injector, described at length in section 3.7. Edits 3, 9 and 10 — arc pressure, the ExaCA thermal-history writer, and the convective wall condition — were deferred, and two of the three have since become necessary.

3. What was built, in the order it happened

3.1 Phase 0 — structure, and a constraint that shaped everything after

Phase 0 established the project layout, a 31-row case matrix generated by a self-auditing script, and one finding that constrains every comparison made since: **melt-pool CFD is chaotic**.

The meaning is specific. The governing equations are deterministic, but the flow they describe amplifies small differences exponentially. Two runs of the same case that differ only in the last bit of one floating-point number — which is what happens when a case is decomposed across a different number of processors, or compiled by a different compiler — diverge to order-unity differences within milliseconds. Measured here: a perturbed *serial* run reached 0.354 relative difference against 0.400 for a parallel one by t = 0.01 s. Parallel decomposition is not a source of error; it is one perturbation among many, and no better or worse than the others.

The consequence is a rule that is enforced mechanically by the case generator: **compare integral and time-averaged quantities, never cell-by-cell values**. Mean temperature between those two runs agreed to 0.44% — energy is conserved, its local distribution is not reproducible. Any claim in this project that rests on two runs agreeing must rest on an integral. This is why, later, a timestep change is checked against mean temperature and deposited mass rather than against peak velocity.

3.2 Phase 1 — two gates on the magnetic machinery

Before building anything, the inherited magnetic module had to be shown to work. Two gates were defined and both passed.

Gate A, the Hartmann benchmark. Hartmann flow is the canonical analytical solution in magnetohydrodynamics: pressure-driven flow of a conducting liquid between two plates with a transverse magnetic field. The field brakes the fast-moving core more than the slow near-wall fluid, flattening the parabolic profile into a plug with thin boundary layers. The analytical velocity profile is

```
u(y)/u(0) = [cosh(Ha) - cosh(Ha*y/L)] / [cosh(Ha) - 1]
```

where the **Hartmann number** Ha = B*L*sqrt(sigma/mu) is the ratio of magnetic to viscous forces. The solver reproduces this to **1.39%**. Two bugs in the checking script had to be fixed first, and both are instructive: Ha was being *fitted* to the data rather than imposed from the sampled field, which slides along the family of Hartmann profiles and can absorb a wrong answer; and the half-width L was taken from the cell-centre span rather than the wall, manufacturing a 16% error at the first cell. A benchmark that fits its own parameter is not a benchmark.

Gate B, coexistence. The magnetic module was run together with VOF, melting, latent heat, Darcy damping and Marangoni on a two-dimensional plate case: 6,906 timesteps, no collapse, magnetic divergence held at 1e-12 across a millionfold conductivity jump between metal and gas. The timestep fell 6.5x but recovered. Switching the field on costs **2.7x** in compute, and every later estimate is multiplied accordingly.

A RESOLVED NULL IS A RESULT

Gate B also compared melt volume with and without a 20 mT field. The difference was **0.14%** against a measured numerical noise floor of 0.4–0.7%. That is a *resolved null*: the effect is smaller than the experiment can see, and the experiment's resolution is quantified. It is a publishable statement.

Melt *velocity*, by contrast, was **unresolved** — the noise floor peaked at 21% and the ordering of the field and no-field runs flips three times during the run. The honest sentence is "below the resolution of this experiment", not "no effect".

This also killed a result already written down. A five-write time average produced a clean "15% braking" that the full time series destroyed. **Check the series, then choose the window, then average** — never the other order.

3.3 Phase 2 — materials, and the number nobody had measured

Zhao's Table 4 gives density, solidus, liquidus, latent heat, surface tension and its temperature coefficient, and cites temperature-dependent specific heat, conductivity and viscosity to a handbook. Those were reconstructed into an OpenFOAM dictionary with the sources chased rather than guessed.

One property the source paper never needed is **electrical conductivity**, $\sigma_e(T)$, because there is no current in their model. It sets both the Lorentz force and the Joule heating here, so it had to be constructed: a solid branch from recommended pure-aluminium values plus a temperature-independent offset fixed by the measured 29% IACS of 5083/5356 alloy, a mushy branch linear in liquid fraction, and a liquid branch. The result runs from 1.64e7 S/m at 300 K to **3.65e6 S/m at the liquidus**.

With a conductivity in hand the force ledger could be computed as stresses rather than as mesh-dependent volumetric source terms. It is reproduced in section 6 because it turned out, months later, to be the explanation for the one measurement that fails. Two other numbers from that analysis: Joule heating is **3.6 W against 1512 W of arc**, or 0.24%, and therefore negligible but now quantified rather than asserted; and the external field beats the arc's own self-field above **13.5 mT**.

3.4 The arc — Goldak's double ellipsoid

An electric arc is not a point heat source. It deposits energy over a region several millimetres across, and the region is not symmetric: the metal ahead of the torch is cold and the metal behind has already been heated, so the isotherms bunch up in front and stretch out behind. Goldak's **double ellipsoid** model, which Zhao uses as his Eq. 27, represents this with two quarter-ellipsoids joined at the arc centre, each a Gaussian in three dimensions:

$$q(x,y,z) = \frac{6\sqrt{3}\, f\, \eta\, U\, I}{a\, b\, c\, \pi\sqrt{\pi}} \exp\left(-\frac{3x^2}{a^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2}\right)$$

with $a = a_f = 3$ mm ahead of the arc centre and $a = a_r = 6$ mm behind, $b = c = 4$ mm, and heat-distribution fractions $f_f = 0.6667$ and $f_r = 1.3333$ summing to 2. Those dimensions are not derived; Zhao obtained them by matching computed and measured pool shapes for his first layer, so they are fitted to this exact process.

Worth noticing: f/a evaluates to **0.2222 for both lobes**. The peak power *density* is identical front and rear. The asymmetry is purely geometric — the rear lobe spreads the same intensity over twice the length. That is what produces the long, shallow-gradient thermal tail behind the arc, and it becomes important in section 6 when the flow turns out to be one-sided.

The arc efficiency eta is 0.7, and Zhao defines it carefully as the efficiency for heating the *molten pool, excluding heating the droplets*. So 0.7 x 16 V x 135 A = **1512 W** is arc power alone, and droplet enthalpy is a genuine addition on top rather than a double count. That definition was assumed for weeks and confirmed from the paper only on 10 September.

3.5 The result that stalled the project: the arc alone cannot melt

With Goldak implemented and everything else off, a bare-plate pass was run for 0.5 s on a graded mesh. The arc delivered 99.97% of its target power steadily across 50,000 timesteps, and mass was conserved to ten digits. The heat source works.

And almost nothing melted. The hot spot saturated at **832 K**. Al–5%Mg has a solidus of 815 K and a liquidus of 906 K, so the arc crosses the solidus by 17 K and falls 74 K short of the liquidus. It produces about one cubic millimetre of mushy skin and no liquid at all, against Zhao's roughly 8 mm wide bead.

Saturation happens because a moving source reaches a quasi-steady state: conduction carries heat away at a rate proportional to the temperature rise, so the rise grows until loss balances input. The timescale is set by diffusion across the source's own size, b^2/α = (4 mm)² / 6.22e-5 m²/s = **257 ms**, which is why half a second was needed to see it.

> **WHY THIS WAS NOT A BUG**
>
> The obvious suspicion was a boundary condition or a missing term. It was neither. In GMAW a large share of the melting energy arrives *with the filler metal*, as superheated droplets at 1506 K, and the model had no filler in it. The arc alone is genuinely not enough — in the real process it is not asked to be.
>
> This moved the wire feed from a Phase 4 convenience onto the critical path. No weld pool could appear before it existed.
>
> An earlier revision of this finding reported 731 K on a smaller uniform mesh with a boundary-condition defect present. **Any document quoting 731 K or "84 K short of the solidus" is superseded.**

3.6 The design decision: place droplets, do not simulate their formation

Zhao's contribution is that he resolves the whole droplet lifecycle — formation at the wire tip, necking, detachment, flight, impingement — coupled to the pool. That is expensive and it is not what this thesis is about. The decision taken was to **inject droplets directly at the interface as spheres with prescribed mass, temperature and impact momentum**, skipping flight entirely.

This is defensible because every parameter of the injected droplet is fixed by the process rather than chosen. The radius follows from wire feed and droplet frequency by conservation of mass (Zhao's Eq. 30):

```
r = cbrt( 3*V_wire*R_wire² / (4*f_droplet) ) = 1.105 mm
```

The frequency is 30 Hz, the temperature 1506 K from Zhao's velocity-inlet condition, and the impact velocity **0.9 m/s** — which is the measured post-detachment velocity from the paper's conclusion (2), *not* the 0.15 m/s wire feed rate. Those differ by a factor of six, and using the wrong one would understate the impingement momentum by 36x in dynamic pressure.

3.7 Edit 4 — the droplet injector, and three failures

The injector accumulates owed volume every timestep and places a sphere when the debt reaches one droplet. It took three attempts, and each failure is worth recording because each was silent.

Background: what a volume-of-fluid solver stores. The metal-argon interface is not tracked as a surface. Instead every cell carries a volume fraction alpha: 1 where the cell is full of metal, 0 where it is full of argon, and intermediate only in the one or two cells the interface passes through. Density, viscosity and conductivity in each cell are blended by alpha. The surface is *defined* as the level set alpha = 0.5, which is why every visualisation in this project contours at that value. Melting is handled separately by the **enthalpy-porosity** method: a liquid fraction epsilon_1 runs from 0 at the solidus to 1 at the liquidus, latent heat is released across that band, and a **Darcy** momentum sink

```
S = -C * (1-epsilon1)² / (epsilon1³ + q) * U
```

drives velocity to zero wherever the metal is solid by treating the mushy zone as a porous medium. That term is unbounded as epsilon_1 goes to zero, which is why solid cells hold velocity at *exactly* zero rather than merely a small value — a fact used repeatedly later as a check.

FAILURE 1 — THE DROPLETS HAD NO EFFECT AT ALL

The injector reported placing 5.5 mm^3 of aluminium at 1506 K, and the temperature field came out **bit-identical** to a run with no droplets. Both cannot be true.

Cause: OpenFOAM's old-time mechanism. A transient solver needs the previous timestep's values to form time derivatives, so every field keeps an `oldTime()` copy. The call that grants write access to a field's interior, `primitiveFieldRef()`, invokes `storeOldTimes()` *before* returning the reference. Writing a value through it therefore snapshots the **pre-injection** state into the old-time level. On the next solve the equation reads that old value back and the injection is erased.

Fix: write alpha, T and epsilon_1 at both time levels.

FAILURE 2 — A FLOATING-POINT EXCEPTION AT THE INJECTION STEP

With the first fix in, the solver aborted with SIGFPE inside a power function at the exact timestep droplet 1 was placed.

Cause: two *other* old-time fields were still argon. The energy equation forms a term in $(\text{rho} - \text{rho}_{\text{old}})/\text{dt}$. With rho = 2650 for the newly placed metal and rho_{old} = 1.6 for the argon that had been there, that evaluates to a heat sink of **9.5e13 W/m^3** acting against an old-time heat capacity of 832. Temperature went hugely negative; the vaporisation model then took the square root of a negative temperature and the trapped-FPE handler aborted the run.

Fix: patch rho.oldTime() and rhoCp.oldTime() to their metal values as well, so that the time derivative correctly vanishes — nothing about that metal is changing between the two levels, it simply is there. Five old-time fields in total.

FAILURE 3 — THE RUN SCRIPT DESTROYED ITS OWN STATE

`Allrun -f` deleted the time directories and then refused to run, because a guard checking for existing logs was evaluated *before* the flag was parsed. The order was fixed in the case builder — but existing case directories keep whatever script they were generated with, and this one still has the old ordering. It cost two benchmark runs on 10 September that silently did nothing and reported the previous run's timings.

The lesson applied since: **never redirect a benchmark's stderr to /dev/null**, and check a log's modification time before trusting a number grepped out of it.

Mass exactness by construction. The injector does not fire on a clock. It accumulates owed volume, and after placing a sphere it subtracts the volume *actually* placed rather than the nominal. Because a sphere placed on a fixed mesh fills a whole number of cells — 44 or 46 or 48 at 0.125 mm^3 each, against a nominal 45.2 — individual droplets are quantised, but the debt carries forward and the next droplet fires earlier or later to repay it. The mean mass flow is exact.

3.8 The 0.1 s run — first liquid metal

Three droplets, full domain, 143,616 cells. `U metal` read **1.4164 m/s** at the first injection. That column had been exactly 0.0000 in every run the project had ever produced — a true zero, because the Darcy source clamps solid cells at machine precision and there had never been any liquid. Mass, energy and momentum all arrived with the droplet.

But there was no melt pool. Six consecutive writes between droplets 1 and 2 showed zero cells above the solidus anywhere in the domain. Each droplet arrived, froze within about 10 ms, and the next landed on cold solid metal 33 ms later. Fusion depth read 0.25 mm, which on a 0.5 mm cell is exactly one cell layer — the mesh talking, not the physics.

The reason the droplets freeze so fast is worth deriving, because it is counter-intuitive and it turns out to be exactly marginal. When two semi-infinite bodies touch, the interface jumps immediately to a temperature weighted by **thermal effusivity** b = sqrt(k*rho*c_p):

```
T_contact = (b1*T1 + b2*T2) / (b1 + b2)
```

Droplet and substrate are the same alloy, so $b_1 = b_2$ = 19,650 and this collapses to the arithmetic mean: (1506 + 300)/2 = **903 K**, which is **three kelvin below the 906 K liquidus**. A droplet of Al–5%Mg landing on a 300 K plate of Al–5%Mg is mushy at the contact plane the instant it touches. And it is fully penetrated at once: the thermal diffusion length sqrt(4*alpha*t) reaches 1.12 mm in 5 ms against a droplet radius of 1.105 mm, so no hot core survives.

The one encouraging signal was a trend. Droplet 1's pool died within 10 ms; droplet 2's was still alive 30 ms later when droplet 3 arrived. The lifetime had more than doubled in the space of two droplets.

3.9 The 0.5 s run — the pool

Fifteen droplets, about three hours on four cores. The prediction going in was that the pool would become continuous, and the reasoning was this. A droplet cools toward whatever the surrounding metal is at, so the time to fall from 1506 K to the solidus is

```
t_life = -tau * ln[ (T_sol - T_inf) / (T_drop - T_inf) ]
```

The numerator shrinks far faster than the denominator as the plate warms, so the lifetime does not creep upward — it **diverges**. At T_{inf} = 300 K it is 0.85 tau; at 700 K, 1.95 tau; at 800 K, 3.85 tau; at the solidus

itself, infinite. Continuity arrives when the lifetime exceeds the 33.3 ms droplet period, well before the background reaches the solidus.

WHAT	RESULT	READING
Mushy continuity	from t = 0.07 s	Last row with zero cells above solidus is t = 0.06. Droplet **2**
Liquid continuity	from t = 0.20 s	Last row with zero cells above liquidus is t = 0.19. Droplet **6**. This is the one that needed the droplets
Sustained flow	never zero after t = 0.06	0.2–0.6 m/s between droplets, 1.2–1.6 at impact. The precondition for the whole magnetic study
Fusion depth	1.75 mm at t = 0.5 s	0.25 (one cell) at droplet 1, then 0.75, 1.25, 1.75 — still climbing at the end
Saturation fit	958 K, tau = 192 ms	Fitted to 13 between-droplet maxima, rms 6.1 K. Clears the liquidus by **52 K**

The fitted time constant of 192 ms sits close to the 257 ms predicted from the Goldak source size, which is why the 0.1 s run — stopped at 39% of it — could never have shown this. The linear power-scaling estimate made in advance was 1009–1071 K against an actual 958 K, an overestimate of 5–12%. That is respectable for a linear extrapolation of a nonlinear system, and the direction and the liquidus crossing were both right.

3.10 What was verified, and what it took to measure honestly

Five independent checks, each of which could have failed.

CHECK	RESULT
Mass	Bead cross-section is fixed before the run by conservation: $A = V_{wire}/v_{travel} = 169.6/20 =$ **8.482 mm^2**. Measured 8.450 — **0.38% low**. 14.95 droplets placed against 15 due. Deposited volume held constant to the digit between injections, bounding the advection error at 2.8e-8 relative
Energy	137.8% of arc energy retained, which is correct rather than a leak: droplets carry their own enthalpy. Bracketed by 133.6% (sensible only) and 145.3% (all latent released); a still-liquid pool means the true value must lie between
Domain	Hottest *metal* cell against the far boundary read **300.3 K** at t = 0.5 s — a 0.3 K rise against a 658 K peak. The far-field boundary condition cannot be influencing anything
Darcy coupling	Velocity is exactly 0.000 through the solidified bead body, over a 6 mm stretch. Not small — zero
Marangoni	Coefficient matches Zhao's Table 4 exactly (-1.55e-4 N/m/K), the sign traces correctly through the momentum equation, and the observed surface flow runs from the arc centre outward as the paper describes

TWO MEASUREMENT DEFECTS THAT WOULD HAVE PRODUCED FALSE RESULTS

Phase contamination. The peak-temperature column maximised over *every* cell, argon included. At one timestep it read 1277.8 K while the hottest metal was 872.5 K — a 400 K error landing squarely on the runaway gate. The same defect made the far-field column appear to rise to 424 K, which looked like a failure of domain independence; it was the argon plume being blown 25 mm sideways at 4–10 m/s. Every extensive maximum is now split by phase.

Threshold versus integral. Deposited mass was measured by summing whole cell volumes wherever alpha > 0.5. That is not a mass; it is the volume of cells that happen to be more than half metal. It appeared to lose 1.5 mm^3 when a droplet flattened and its interface smeared across more cells. The conserved quantity is the integral of alpha times cell volume, and measured that way the mass is constant to eight digits.

3.11 The measurement that fails

With the pool established, the bead could finally be compared against Zhao's Table 5. Slicing the alpha = 0.5 surface along the bead centreline gives the free-surface profile as a single line, and its crown height is **2.60 mm**. Zhao reports **1.55 mm simulated and 1.80 mm measured** at the equivalent mid-pass location. The replication gate is 15%; this is **+68%**.

Because the cross-sectional area is pinned by mass conservation, height determines width. For a parabolic cap, w = 3A/2h, so a 2.60 mm crown implies a **4.9 mm** wide bead where Zhao's is **8.2 mm**. The bead is a narrow hump where it should be a broad ridge. It is not spreading. Section 6 diagnoses why.

3.12 Infrastructure closed out

Two items open since Phase 1 were closed on 9–10 September. The work now lives in a GitHub repository rather than in one place on a virtual disk, with an ignore policy that keeps run output out — five case directories holding gigabytes on disk contributed 62 KB to the push, because everything except `initial/`, `constant/`, `system/` and the run script is regenerable. That also means a new machine needs a `git clone` and no file transfer at all.

4. The droplet stream, examined

A question raised on inspection of the animation was whether droplets were landing behind their predecessors or on top of them. The injector logs every placement, so this can be answered exactly rather than by eye.

The landing point tracks the moving arc: $x_c = x_0 + v*t$ evaluated at the injection instant. With the arc starting at x = 0 and travelling at 20 mm/s, the fifteen droplets of the 0.5 s run landed at 0.666, 1.315, 1.993, 2.641, 3.260, 3.967, 4.616, 5.264, 5.971, 6.620, 7.268, 7.976, 8.624, 9.272 and 9.980 mm, at times from 0.0333 s to 0.4990 s. **Every step is forward. None repeats.**

WHY IT LOOKS AS THOUGH THEY LAND IN THE SAME PLACE

Mean spacing is **0.665 mm**, which is exactly v/f = 20/30.02. The droplet is **2.21 mm** across. Consecutive droplets therefore overlap by **70%** — and that is correct physics, not an artefact. A 70% overlap is how a continuous bead forms rather than a row of separate ones. Zhao's process has the identical 30 Hz at 20 mm/s.

On a 0.5 mm mesh it is sharper still: the footprint is **4.4 cells** wide and advances **1.33 cells**. Because 1.33 is not an integer, the voxelised footprint advances one cell for two injections and then two cells on the third, so the filled cell set genuinely does repeat.

That 1.33-cell beat is visible in the mass record. The cells-placed sequence is 44, 46, 44, 42, **48**, 44, 44, **48**, 44, 44, **48**, 44, 44, **48**, 44 — the 48s fall on droplets 5, 8, 11 and 14, a period of exactly three, because 3 x 1.33 = 4.0 cells realigns the sphere with the lattice.

The release height, and a hypothesis that was refuted. The injector places each sphere clear of existing metal by searching for the highest metal within one droplet radius of the arc axis and adding a standoff. Since the search radius (1.105 mm) exceeds the droplet spacing (0.665 mm), that search always finds the previous droplet's material — so every droplet is released above the last one's crown. The concern was that this would stack deposits vertically and contribute to the tall bead.

The log refutes it. Release height goes 1.355, 1.355, 1.855, 1.855, 2.355 and then **stays at 2.355 mm for ten consecutive droplets**. It climbs for four droplets while the bead builds from nothing and then saturates. The injector is not stacking, and that candidate explanation is dead.

The three values decode to a free surface at -0.25, +0.25 and +0.75 mm — consecutive cell centres, because the search returns a cell centre rather than the interpolated surface. The apparent conflict with a 2.60 mm bead crown is not one: the search cylinder sits over the *arc*, at the leading edge, where only part of the material that will eventually accumulate there has arrived. Each position receives about 3.3 droplets over roughly 110 ms as the arc passes; the full-height bead is behind.

Droplet resolution is the one genuine artefact. The sphere is 2.21 mm across on a 0.5 mm mesh — **4.42 cells**, where volume-of-fluid practice wants ten to twenty. Worse, the injector fills whole cells: a cell centre is either inside the sphere or not, with no partial filling, so the droplet is a voxel staircase rather than a ball. Its surface area is inflated by up to 50% (the axis-aligned staircase limit is 3/2 the true area), and it meets the pool on a flat face several cells wide rather than at a point. Surface tension cannot smooth this away, because the Continuum Surface Force model computes surface tension from the gradient of alpha *on the same mesh* — it cannot represent structure the mesh cannot hold.

5. The diagnosis: why the bead is too tall

A 2.60 mm bead where Zhao gets 1.55 mm is a 68% error in the quantity the replication gate is written about. Four candidate explanations were tested and three were eliminated.

CANDIDATE	VERDICT
Marangoni sign or magnitude wrong	**Eliminated.** The coefficient in the case dictionary matches Zhao's Table 4 exactly at -1.55e-4 N/m/K; the sign traces correctly through `UEqn.H`, where a negative term in the matrix assembly becomes a positive source; and the measured surface flow runs outward from the arc as the paper describes
The injector stacks droplets vertically	**Eliminated.** Release height saturates after four droplets and holds for the remaining ten
Droplets land in the wrong place	**Eliminated.** Every landing advances by 0.665 mm, exactly v/f
Forces missing from the momentum balance	**Confirmed.** See below

The force ledger computed in Phase 2, expressed as stresses over the pool scale so that the comparison is mesh-independent, ranks the six surface and body forces acting on the molten pool. Set against what the model actually contains:

FORCE	Pa	IN THE MODEL?	WHAT IT DOES
Self-field electromagnetic	**435**	**NO** — Phase 6	The welding current crossing its own magnetic field. Drives flow downward and outward, increasing penetration and width
Surface tension	283	yes	Pulls the free surface toward minimum area. **Domes the bead**
Arc pressure	**228**	**NO** — Phase 3, deferred	The arc plasma jet pressing on the pool surface. Flattens it and produces the depression Zhao reports
Gravity	78	yes	Flattens, but only weakly at this scale
Marangoni	15.5	yes	Surface shear from the temperature dependence of surface tension. Sets the direction of the surface circulation rather than the bead shape
Buoyancy	0.5	yes	Negligible, now quantified

THE ARITHMETIC OF A BALLED-UP BEAD

Surface tension domes the bead at **283 Pa**. The only force in the model opposing it is gravity at **78 Pa** — a 3.6:1 loss. A narrow, tall deposit is exactly what that imbalance predicts.

The capillary pressure holding a 2.6 mm bead domed is 2*sigma/R = 2(0.87)/2.6e-3 = **670 Pa**; hydrostatic is only rho*g*h = **68 Pa**. Gravity cannot flatten this bead by an order of magnitude.

Zhao has **663 Pa** of additional force that this model does not: his momentum source, Eq. 21, reads $F = F_g + F_{buo} + F_{emm} + F_{st} + F_{Ma} + F_{arc}$, and his Table 4 even gives the arc-pressure distribution radius, 2 mm.

This was predicted on paper in Phase 2 and measured in September. The bead-shape failure is independent confirmation of a scaling analysis done weeks earlier — good news for the analysis, bad news for the current model.

5.2 The missing crater, and why it is the same problem

Zhao's conclusion (2) reports that the droplet falling at 0.9 m/s produces "an expansion of molten pool size and a depressed region formed in the molten pool". The measured free-surface profile here is a smooth mound with no dip.

Two honest caveats before treating that as a discrepancy. First, Zhao's depression is **transient**: his Fig. 10 tracks it forming and then vanishing "under the action of surface tension" within about 24 ms, which is less than the 33.3 ms droplet period — so a profile taken between droplets would show nothing even in his model. Second, at 0.5 mm cells any depression this model produced would be quantised to one cell.

But the same force ledger explains it. A depression is made by pushing the surface down, and the two forces that push are the two that are absent. There is a second-order contribution too: the voxelised droplet meets the pool on a flat face several cells wide rather than at a point, spreading its impulse over a larger contact area and blunting exactly the concentration that would dent the surface.

6. The plan

6.1 Where the work sits against the roadmap

PHASE	PLANNED	ACTUAL
P0	structure, case matrix, chaotic-sensitivity constraint	**complete**
P1	toolchain, Gates A and B	**complete.** The two items left open — the GitHub remote and HPC access — are now one closed and one in progress
P2 (wk 2)	materials, sigma$_e$(T), force scaling	**complete.** The roadmap still lists this as "next"
P3 (wk 3–5)	Goldak, **arc pressure**, **surface heat loss**	**one third done.** Goldak is in and verified positioned to the millimetre. Arc pressure and heat loss are the immediate next work
P4 (wk 5–7)	wire feed; gate on droplet radius, frequency, velocity	**gate passed** by a different route. Radius 1.105 mm exact, frequency 30.02 Hz measured, impact velocity 0.9 m/s prescribed
P5 (wk 7–10)	replication within 15% of Table 5. **Submittable thesis from here**	**first attempt made, fails at +68%.** This is the gate that matters
P6 (wk 10–12)	welding-current injection. *The novel result*	not started. Supplies the missing 435 Pa
P7 (wk 12–14)	13-case magnetic matrix	not started. Needs a server
P8–P9	microstructure, verification, writing	not started

The roadmap is dated 1 September and this document is dated the 10th. In nine days P2 has closed, most of P3 and P4's gate have been done, and P5 has had a first attempt. That is ahead of a twenty-week plan — but P5's gate is the one that decides whether there is a submittable thesis, and it has not been met.

6.2 Immediate — four steps, none of which need a server

#	STEP	WHY NOW
1	**Edit 3 — arc pressure.** Zhao Eqs. 25–26, distribution radius 2 mm from his Table 4	Completes P3 and is the highest-value change available. Supplies 228 Pa of the 663 Pa deficit and addresses the bead height and the missing depression with one term
2	**Edit 10 — surface heat loss.** Zhao Eq. 29, convective plus radiative, h = 80 W/m^2/K, on **all** wall patches	Also P3. The substrate underside reached 557.7 K by t = 0.5 s and would reach 700–800 K by 2 s, where an adiabatic wall traps about 7% of the input — enough to distort the exact dimension the P5 gate measures. On all walls rather than just the bottom, because at these durations the thermal field reaches every boundary
3	Move the arc start to x = -10 mm	The arc currently begins at x = 0 in a domain running -25 to +35 mm, so the heat source's leading edge reaches the far boundary at **t = 1.60 s**. Starting further back buys 9 mm of clearance behind and 12 mm ahead
4	Run **1.5 s**, about 9.3 hours on the desktop	Not 2 s. The thermal diffusion length sqrt(4*alpha*t) reaches 22.3 mm at 2 s against 25 mm to the side boundary — the domain-independence result earned at 0.5 s (ratio 2.24) does not survive to a ratio of 1.12

The measurement that closes the loop is the bead crown taken mid-pass, away from the arc-on and arc-off transients, against Zhao's 1.55 mm simulated and 1.80 mm measured — which he also takes mid-pass, at X = 25 mm of a 43 mm deposit. Like for like.

6.3 Queued behind that

ITEM	RATIONALE
Fractional-alpha droplet placement	Replace the binary in-or-out cell test with a subsampled volume fraction, and energy-weight the temperature so a partly-filled cell does not receive a full droplet's enthalpy. Removes the voxel staircase and, as a side-effect, the 44/46/48-cell mass quantisation. About twenty lines. It does not fix the resolution — a 4.4-cell sphere stays coarse — but coarse-and-smooth is a legitimate VOF representation and coarse-and-staircased is not
Regression on Gates A and B	Both were last run against the solver at Edit 8. The binary has changed since. The droplet code is inert without a `droplet_radius_m` entry, so the gate cases should be unchanged — which is exactly the sort of assumption that deserves a five-minute check rather than trust
Mesh refinement study	Eight cells per droplet diameter needs 0.276 mm cells and 0.85 M cells. That is 43k cells per rank on twenty cores — efficient — and 7,180 per rank is what the current mesh would give, which is too thin to scale. This is the run the shared server is actually for
P6 — welding-current injection	Route A first: impose the azimuthal field from Ampere's law, $H = I_{enc}(r)/2*pi*r$, as a boundary profile and let the induction equation carry it inward. Boundary conditions only, no solver surgery. Validate at $B_{ext} = 0$ against Zhao's analytic force fields, Eqs. 22–24, before any external field is applied

6.4 Compute

The desktop — six cores, sixteen gigabytes — runs the current mesh at 22,340 seconds of wall clock per simulated second on four ranks, which is where the 9.3 hour figure comes from. A benchmark showed no benefit in going to six ranks, and two concurrent runs launched by accident halved the rate, which is the clearest possible demonstration that this machine is saturated.

The shared university server — twenty cores, sixty-four gigabytes, Windows 10 — is currently occupied by another group. When it becomes available it is worth having for the refined mesh and for P7's thirteen magnetic cases, not for the immediate runs. Its sixty-four gigabytes matter chiefly for post-processing: ParaView on the desktop has been running at 90% of its available memory, which is a WSL default allocation rather than a hardware limit and can be raised.

7. Reference

7.1 The numbers that matter

QUANTITY	VALUE	SOURCE
Arc power	1512 W	0.7 x 16 V x 135 A; efficiency excludes droplets
Droplet stream power	679 W	measured deposit, sensible plus latent
Arc-alone saturation	832 K	measured; +17 K on solidus, -74 K on liquidus
Saturation with droplets	958 K	fitted, 13 points, rms 6.1 K
Solidus / liquidus	815 / 906 K	Zhao Table 4
Droplet radius / frequency	1.105 mm / 30.02 Hz	Zhao Eq. 30 from wire feed
Droplet spacing	0.665 mm	v/f; 70% overlap
Impact velocity	0.9 m/s	Zhao conclusion (2), post-detachment
Bead cross-section target	8.482 mm^2	V_{wire}/v_{travel}, fixed before any run
Bead cross-section measured	8.450 mm^2	-0.38%
Bead crown	2.60 mm	vs 1.55 / 1.80 mm. **The failing number**
Thermal diffusivity	6.22e-5 m^2/s	155 / (2650 x 940)

QUANTITY	VALUE	SOURCE
Contact temperature	903 K	$(T_{drop}+T_{plate})/2$, equal effusivities
Arc saturation timescale	257 ms	b^2/alpha, Goldak b = 4 mm

7.2 The measurement tools

Two scripts do all the quantitative reading, and both have been revised after producing a wrong answer. `pool_stats_waam.py` (rev. 4) reports peak temperature, wall and far-field temperatures, pool size and depth, velocity and the energy balance, splitting every extensive maximum by phase. `check_droplet.py` (rev. 2) reports deposited mass as the integral of alpha, self-calibrating the substrate from the initial time rather than assuming it. Both refuse to run without OpenFOAM's own cell centres and volumes rather than assuming a uniform mesh — a deliberate refusal, because three separate wrong results in this project came from a script quietly assuming a value it could not determine.

THE THREE RULES THESE FAILURES PRODUCED

Split every measurement by phase. Argon is hotter and an order of magnitude faster than the metal, and reporting either as a pool quantity is wrong in the flattering direction.

Measure the conserved quantity, not a thresholded proxy. Sum alpha times volume; never count cells above a cutoff.

Never let a script assume what it can read. And never redirect a diagnostic's stderr to /dev/null — two benchmark runs on 10 September reported a plausible number from a run that had never started.